

There are seven problems in total. You must solve the first six. Problem 7 is optional.

Problem 1 (CO5): Pumping Lemma (5 points)

Consider the following language.

$$L = \{0^i 1^j 2^k : i, j, k \geq 0 \text{ and if } i = 1 \text{ then } j = k, \text{ otherwise } j \neq k\}$$

Prove that L is not regular using the pumping lemma.

Problem 2 (CO3): Context-Free Grammars (10 points)

Consider the following languages.

$$L_1 = \{0^i 1^j 2^k : i, j, k \geq 0 \text{ and } i > j + k\}$$

$$L_2 = \{w \in \{0, 1\}^* : w \text{ contains 0 in every second and third position}\}$$

$$L_3 = \{w_1 \# w_2 : w_1, w_2 \in \{0, 1\}^* \text{ and the number of 0s in } w_1 \text{ is twice the number of 1s in } w_2\}$$

$$L_4 = \{w \# w^R \in L_3 : w \in L_2\}$$

Now solve the following problems.

- (a) **Give** a context-free grammar that generates L_1 . (2 points)
- (b) **Give** a context-free grammar that generates L_2 . (2 points)
- (c) **Write** four strings of length ten in L_3 that end with $0\#0101$. (1 point)
- (d) **Give** a context-free grammar that generates L_3 . (2 points)
- (e) **Write** all strings of length less than eight in L_4 . [Note: w^R denotes the reverse of w , for example, if $w = 01101$, then $w^R = 10110$.] (1 point)
- (f) **Give** a context-free grammar that generates L_4 . (2 points)

Problem 3 (CO4): Turing Machines (10 points)

- (a) Given the Turing machine configuration $000q_211$, identify the tape contents, the current state, and the current head position of the machine. (2 points)

Consider the following languages.

$$L_1 = \{w \in \{0, 1\}^* : \text{Every 0 in } w \text{ is followed by at least one 1}\}$$

$$L_2 = \{w \in \{0, 1\}^* : w \text{ contains equal number of 0s and 1s}\}$$

$$L_3 = \{\langle D, w \rangle : D \text{ is a DFA that accepts } w \text{ and } w \text{ contains even number of 1s}\}$$

Now solve the following problems.

- (b) **Give** the state diagram of a Turing machine that decides L_1 . (3 points)
- (c) **Give** the state diagram of a Turing machine that decides L_2 . (3 points)
- (d) **Give** a Turing machine that decides L_3 . (2 points)

Problem 4 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Consider the following context-free grammar.

$$S \rightarrow PbQ$$

$$P \rightarrow aP \mid bP \mid \varepsilon$$

$$Q \rightarrow aQa \mid aQb \mid bQa \mid bQb \mid \varepsilon$$

- (a) **Give** a leftmost derivation for the string 'abaabbab'. (3 points)
- (b) **Draw** the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) **Show** that the given grammar is ambiguous by drawing two more parse trees (apart from the one that you have drawn in (b)) for the given string in (a). (4 points)
- (d) **Write** two strings of length four that have exactly one parse tree in the given grammar. (1 point)

Problem 5 (CO3): Non-Regular Languages (5 points)

Write whether the following statements are true or false. Write also the correct statements for the false statements.

- (a) Regular language $\subset$ context-free language $\subset$ recognizable language $\subset$ decidable language. (1 point)
- (b) Some non-regular languages may satisfy the pumping lemma. (1 point)
- (c) For recognizable languages, the Turing machine halts on every input. (1 point)
- (d) Pushdown Automata can recognize every regular and non-regular language. (1 point)
- (e) A context-free grammar for a regular language cannot be ambiguous. (1 point)

Problem 6 (CO3): Pushdown Automata (10 points)

Consider the following languages.

$$L_1 = \{w \in \{0,1\}^* : w \text{ ends with } 01\}$$

$$L_2 = \{w_1\#w_2 : w_1, w_2 \in \{0,1\}^*, 3|w_1| = |w_2|\}$$

$$L_3 = \{w\#w^R : w \in \{0,1\}^* \text{ and } w \text{ contains an odd number of } 0\text{s}\}$$

- (a) **Give** the state diagram of a PDA that recognizes L_1 . (3 points)
- (b) **Give** the state diagram of a PDA that recognizes L_2 . [Recall: $|w|$ denotes the length of w .] (3 points)
- (c) **Give** the state diagram of a PDA that recognizes L_3 . (4 points)

Problem 7 (Bonus): (5 points)

Let $L = \{\langle D \rangle : D \text{ is a DFA such that if any string } w \text{ is accepted by } D, \text{ it contains at least one } 1\}$.
Prove that L is decidable.

After you are finished with the exam, please indicate where you stand on the smiley face spectrum.

